

SHOVAN SHAKYA

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🐙 [momoisgoodforhealth.github.io/](https://github.com/momoisgoodforhealth)

Education

University of South Florida

Bachelors of Science in Electrical Engineering

Tampa, Florida

May 2025

Skills

Languages: C, C++, Python, Verilog

Software Development Tools: Zephyr RTOS, OpenCV, CUDA, Vivado, ROS2, CMake, Git

CAD Tools: Altium, KiCAD

Hardware: Oscilloscope, Logic Analyzer

Experience

REEKON Tools

Boston, Massachusetts

Embedded Software Engineer

June 2025 - Current

- Primary firmware developer for the T1R Range hybrid measuring tool (nRF54 + Zephyr RTOS), extending an existing C based production codebase to integrate new LDM and IMU subsystems, add IPC interfaces between Arm and RISC-V cores, and advance the system from prototype to production readiness supporting \$300K+ Kickstarter pre-orders.
- Developed an Embedded Linux + STM32 based LDM scanning prototype integrating camera sensors, stepper motors and quadrature encoder feedback for planar measurements.
- Implemented peripheral drivers (UART for LDM, stepper motors, I2C for battery charger, GPIO expander, magnetic sensor, IMU) and expanded BLE capabilities.
- Developed multithreaded C++ image processing and control pipelines to detect laser dots, steer to target pixels, and synchronize encoder and distance measurements for planar surface reconstruction.
- Improved SPI display driver refresh rate from 8 Hz to 30 Hz using semaphore based synchronization for tear-free LCD updates; implemented adaptive brightness control via ALS feedback and tuned display voltage rails to reduce power consumption while maintaining contrast and readability.
- Implemented dual-axis leveling from accelerometer data with 30 Hz update rate and LPF for stable visualization.
- Implemented CUDA accelerated Goertzel filtering across pixels to isolate LED blink frequencies in video streams, increasing blob tracking throughput from 5 Hz (CPU) to 40 Hz (GPU) for prototype testing.

Jaycon Systems

Palm Bay, Florida

Firmware Engineering Intern

September 2024 - May 2025

- Enhanced STM32 production firmware in C by mitigating SPI EMI issues for FCC compliance, adding ADC based temperature logging, and implementing SPI double buffering to eliminate DMA buffer corruption.
- Exposed sensor data (thermistor, VIN, GPIO) to a desktop application via C++ with SWIG generated C# bindings for diagnostics and monitoring.
- Contributed to a Zynq-7000 SoC image processing and state estimation system for a NASA SBIR project, implementing connected component labeling (CCL) in C++ on ARM Cortex-A9 (15 Hz at 720p) and developing FPGA-accelerated pipelines using C++ HLS and SystemVerilog with dual-port BRAM integration.

NBCUniversal

Orlando, Florida

Sensor Fusion Intern

May 2024 - August 2024

- Developed STM32 firmware in C for IMU interfacing over I2C, communicating with other STM32 modules via CANBUS, and transfer IMU data to a computer using UDP via Ethernet SPI.
- Designed a 4-layer STM32-based PCB (2cm x 2cm) with BNO088 IMU, Ethernet SPI module and TJA1051 CAN IC, supporting up to 30V input voltage.
- Developed a Python tool for synchronized pose and camera data capture on a NVIDIA Jetson based ROS2 platform.
- Applied image processing techniques to detect cutouts on display screens to detect bugs and verify manufacture quality.

Monterey Bay Aquarium Research Institute

Moss Landing, California

Computer Vision Intern

June 2023 - August 2023

- Developed OpenCV based realtime disparity and distance estimation tool for fisheye stereo cameras with accuracy range of 5 meters (+- 300 mm) for desktop and VR.
- Used multiprocessing to perform disparity and tracking in parallel.
- Integrated FathomNet YOLOv5 deep sea organism tracking model with distance estimation.
- Used Sockets to send realtime frames, and distance information between Python OpenCV application and Unity VR.